ISO 230-2 linear-axis identification

A4 SETTINGS SHEET

Validated v3 settings and changes from the first run

FIXED OPERATING POINT

2.0 mm/s | 40 mm/s^2 | 2.0 s settle + 0.5 s record

5 runs / 240 stops
240/240 arrivals resolved

DRIVE, MOTOR AND HOMING

Screw lead / motor	**2.0 mm / 200 steps rev^-1**
Microstep mode	**MRES 16; 0.625 um microstep**
Run / hold current	**400 mA / IHOLD = IRUN**
Driver mode	**SpreadCycle; intpol = 0**
Encoder	**not fitted; open loop**
ISO velocity	**2.0 mm/s**
ISO acceleration	**40.0 mm/s^2**
Pulse rate at ISO speed	**3,200 Hz**
Home side	**minimum switch**
Home seek / slow speed	**2.0 / 0.15 mm/s**
Home backoff	**2.0 mm**
Measure far end	**enabled**

TRAVEL, TARGETS AND SEQUENCE

Measured range of motion	**122.695 mm**
Software limits	**2.000 to 120.695 mm**
Reference position x_ref	**61.350 mm**
Formal target span	**5.000 to 117.500 mm**
Number of targets	**24**
Mean target interval	**4.891304 mm**
Nominal spacing / seed	**5 mm / 2302**
Overrun / ref. preapproach	**3.0 / 3.0 mm**
Datum / end-reference visits	**3 / 3**
Formal ISO runs / stops	**5 / 240**
Approx. one-run duration	**5.05 min**
Approx. complete duration	**25.81 min**

STOP TIMING AND AUTOMATIC STARTUP

Settle	Record	Total dwell	Boot countdown	Centre wait	End state
2.0 s	**0.5 s**	**2.5 s**	**10 s**	**15 s**	**motor held**

DIFFERENCES FROM THE FIRST RUN

Parameter	First run	Fixed v3	Reason / consequence
ISO velocity	5.0 mm/s	**2.0 mm/s**	Keeps recorded response below the observed ~2.5 mm/s limit
Pulse rate	8,000 Hz	**3,200 Hz**	Derived directly from the lower velocity
Settle before record	0.5 s	**2.0 s**	Provides a clearly stationary plateau before sampling
Total stop dwell	1.0 s	**2.5 s**	2.0 s settle + unchanged 0.5 s record
Automatic centre wait	0 s	**15 s**	Allows acquisition to be running before the ISO sequence
One-run duration	2.113 min	**5.05 min**	Longer motion and dwell; time was not the priority
Five-run duration	10.809 min	**25.81 min**	Derived complete ISO sequence duration

HELD CONSTANT

- acceleration: 40 mm/s^2
- record window: 0.5 s
- run current: 400 mA; MRES 16
- homing speeds, backoff and far-end measurement
- target table, seed, overruns and reference geometry
- five runs, 24 targets and 240 formal stops

INTERPRETATION AND VALIDATION

The diagnostic runs reached the intended endpoints. The change therefore addressed timing and availability of a stable recorded plateau - not a demonstrated permanent positional shortfall.

v3 result: 251/251 planned plateaus paired; 240/240 ISO stops; arrival error median -0.02 um, RMS 1.63 um, max |error| 4.77 um.

TRACEABILITY first-run config 96d325dacb94... | fixed config 694408ce62ee... | target: iso_targets_seed2302_rom122.695.csv

Evidence boundary: no matching terminal log was saved for the diagnostic or v3 acquisition. The data prove endpoint completion and the recorded time response, but do not by themselves isolate whether the ~2.5 mm/s ceiling originated in motion execution or the metrology/output path.

Source: iso_config.json and generated iso_plan_summary.json | Prepared 22 October 2026